Beauty and the Labor Market

An Econometric Replication and Extension

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Abstract

Does the labor market reward above-average appearance, penalize below-average appearance, or both? We examine this question using 1,260 worker observations from the historical beauty dataset associated with Hamermesh and Biddle (1994). Ordinary least squares estimates reproduce a negative association between below-average appearance and log hourly wages after adjustment for observed worker and job characteristics. The coefficient of −0.133 implies a 12.4 percent wage difference, with a 95 percent confidence interval from −19.0 to −5.4 percent. The estimated premium for above-average appearance is 1.2 percent, but it is imprecise. Conditional quantile estimates are more negative at the upper end of the wage distribution, but a direct 90th-versus-10th-percentile comparison is not statistically significant ($p = 0.225$). A correctly specified gender interaction model yields nearly identical below-average appearance penalties for men and women. The analysis supports an asymmetric conditional wage association in this historical sample; it does not identify a causal effect of appearance or establish persistence in contemporary labor markets.

Keywords: Physical appearance, wage inequality, conditional quantile regression, replication

Scope: Historical cross-sectional evidence; all reported estimates have been recomputed from a public copy of the teaching dataset.

Introduction

Wage differences are usually interpreted through education, work experience, and the demands of a job. Physical appearance raises a more difficult question: how much do earnings also reflect characteristics that employers or customers value but that standard measures of human capital do not capture? An appearance-related wage gap matters economically because it may reflect unequal treatment, differences in job allocation, or interpersonal skills that conventional wage equations measure poorly. These explanations carry different implications, even when they generate similar regression coefficients.

This paper reassesses the central finding of Hamermesh and Biddle (1994): workers rated below average in appearance earn less than otherwise observable-similar workers rated average, while the advantage associated with above-average appearance is smaller. We reproduce the baseline wage equation, examine sensitivity to controls and influential wages, and extend the analysis to conditional wage quantiles and gender interactions. The purpose is to establish which conclusions are supported by the available data and which require a stronger research design.

Economic context and research question

Hamermesh and Biddle (1994) analyze interviewer assessments of appearance in three surveys conducted in the United States and Canada in 1971, 1977, and 1981. They report a below-average appearance penalty of roughly 5–10 percent, generally larger than the premium for above-average appearance. Their findings also connect appearance to occupational sorting and labor-force participation. This is historical evidence, and reproducing it does not test whether appearance-related wage differences have remained unchanged since the original study.

Several mechanisms could produce a conditional wage gap. Employers may prefer workers with certain appearances, or customers may respond differently to them. Employers may also infer ability from appearance even when appearance is uninformative about the task. Alternatively, appearance may be correlated with communication skills, health, family resources, or confidence.

Möbius and Rosenblat (2006), using an experimental labor market, distinguish confidence, employer perceptions of ability, and oral interaction as channels through which attractiveness can influence wage offers. Their setting demonstrates why an observed wage premium alone cannot establish a single mechanism.

Our empirical question is therefore narrower than whether employers discriminate based on appearance. We ask whether wages differ across interviewer-rated appearance categories after adjustment for the characteristics available in this dataset. We then ask whether the size of that association varies across conditional wage quantiles or between men and women. These extensions can describe heterogeneity, but they do not by themselves separate employer preferences, customer responses, and unmeasured productivity.

Data and measurement

The analysis uses the public Wooldridge beauty teaching dataset, containing 1,260 observations and 17 original variables. The package documentation identifies Hamermesh and Biddle (1994) as the source and explains that only a subset of the original variables is retained (Shea, n.d.). Consequently, this exercise partially replicates the study using a teaching extract, rather than reproducing every sample and specification in the original article. The extract does not retain survey identifiers or sampling-design variables sufficient to reconstruct the original surveys separately. We do not assign these observations a new collection date.

The public copy was checked against the original baseline output, reproducing the appearance coefficients and R^2 to the reported precision. All 1,260 observations have complete values for the variables used here. No observations are dropped from the main regressions, and the reported estimates are unweighted.

Appearance is recorded on a five-point interviewer scale. We group ratings 1–2 as below average, rating 3 as average, and ratings 4–5 as above average. Average appearance is the reference category throughout. These labels describe recorded judgments; they are not objective measures of

personal worth or productivity. External ratings avoid self-assessment, but they can still reflect interviewer preferences, social norms, and characteristics observed during the interview.

Table 1: *Descriptive statistics*

Variable	Mean	SD	Range
Hourly wage	6.307	4.661	1.02–77.72
Log hourly wage	1.659	0.595	0.02–4.35
Years of education	12.563	2.624	5.00–17.00
Years of experience	18.206	11.963	0.00–48.00
Below-average appearance	0.123	0.329	0.00–1.00
Above-average appearance	0.304	0.460	0.00–1.00
Female indicator	0.346	0.476	0.00–1.00
Black indicator	0.074	0.262	0.00–1.00
Union member	0.272	0.445	0.00–1.00
Good health	0.933	0.250	0.00–1.00
Married	0.691	0.462	0.00–1.00
South	0.175	0.380	0.00–1.00
Big city	0.219	0.414	0.00–1.00
Small city	0.467	0.499	0.00–1.00
Service industry	0.274	0.446	0.00–1.00

Source. Authors' calculations from the public beauty dataset. N = 1,260 for every variable. Indicator means are sample proportions. Wages are recorded in the dataset's historical units and are not converted into current purchasing power. Experience squared is included in the regressions but omitted from this descriptive table.

Mean hourly wages are 6.31 and the median is 5.30, while the maximum is 77.72. The natural logarithm reduces the influence of the long upper tail and allows wage differences to be expressed proportionally. It does not remove heteroskedasticity or guarantee a correctly specified conditional mean. We therefore use heteroskedasticity-robust inference and examine sensitivity to the highest wages.

Below-average appearance accounts for 155 observations, or 12.3 percent of the sample. Average and above-average appearance account for 722 observations (57.3 percent) and 383 observations (30.4 percent), respectively. The sample includes 824 men and 436 women. The below-average groups contain 96 men and 59 women, which limits precision when estimating gender differences. The proportion living in the South is 17.5 percent.

Education and experience capture central dimensions of measured human capital. The remaining variables describe health, demographic characteristics, union membership, marital status, region, city size, and service-industry employment. The female and Black indicators follow the source coding and cannot represent the full complexity of gender or race. A single service-industry indicator also provides only a coarse measure of the work people perform.

Empirical strategy

The baseline ordinary least squares model relates log hourly wages to two appearance indicators and observed controls. We write it as

$$\ln(w_i) = \alpha + \beta_B B_i + \beta_A A_i + \mathbf{X}_i'\gamma + \varepsilon_i$$

Here, B indicates below-average appearance, A indicates above-average appearance, and X contains the controls. We report three nested specifications. Model 1 contains only the two appearance indicators. Model 2 adds education, experience, experience squared, the female and Black indicators, and good health. Model 3 additionally includes union membership, marital status, the South indicator, city size, and service-industry employment. Holding the sample constant makes changes in coefficients attributable to specification rather than changes in sample composition.

The coefficients on B and A compare each group with average-rated workers at the same included covariate values. A log coefficient b corresponds to a proportional difference of 100 × [exp(b) − 1] percent. We apply the same transformation to confidence-interval endpoints. This is an exact transformation of the fitted log-wage contrast; interpreting it as a difference in expected wage levels requires assumptions about the residual distribution. In particular, exponentiating a fitted log wage alone does not recover the conditional arithmetic mean wage.

OLS inference uses HC1 heteroskedasticity-robust standard errors and t-based confidence intervals. We test the two appearance coefficients jointly. We also test whether the penalty and premium have equal magnitudes and opposite signs, using the restriction βB + βA = 0. A significant penalty and an insignificant premium do not, on their own, establish a statistically significant asymmetry.

We use conditional quantile regression to investigate differences beyond the conditional mean:

$$Q_\tau[\ln(w_i) \mid B_i, A_i, \mathbf{X}_i] = \alpha_\tau + \beta_{B,\tau} B_i + \beta_{A,\tau} A_i + \mathbf{X}_i' \gamma_\tau$$

The quantiles are 0.10, 0.25, 0.50, 0.75, and 0.90, with the full Model 3 controls included at each quantile. Inference uses 499 paired nonparametric bootstrap replications with seed 20261105. The same resampled observations are used across quantiles, allowing covariance between quantile estimates to enter the test comparing the 90th and 10th percentile coefficients. Displayed intervals are pointwise normal-approximation intervals, not simultaneous confidence bands.

The gender extension includes the female main effect and its interactions with both appearance indicators. Men's appearance contrasts are βB and βA; women's are βB + θB and βA + θA. Their standard errors use the full covariance matrix. We test the interaction terms jointly rather than inferring a gender difference from the significance of separate coefficients.

These estimators describe conditional associations. A causal interpretation would require that appearance be unrelated to omitted wage determinants after conditioning on X, an assumption the dataset cannot establish. Some controls may also lie on a pathway from appearance to wages: job

sorting, for example, could make service-industry employment a mediator rather than a conventional confounder. We therefore compare the full model with one omitting this variable.

Mean wage differences

Table 2: *Appearance and log hourly wages*

	Model 1	**Model 2**	**Model 3**
Below-average appearance	-0.2088 (0.0515)	-0.1574 (0.0411)	-0.1326 (0.0395)
Above-average appearance	-0.0454 (0.0376)	-0.0081 (0.0314)	0.0122 (0.0307)
Education	—	0.0652 (0.0055)	0.0673 (0.0056)
Experience	—	0.0406 (0.0042)	0.0407 (0.0042)
Experience squared	—	-0.0006 (0.0001)	-0.0006 (0.0001)
Worker controls	No	Yes	Yes
Additional job and location controls	No	No	Yes
Observations	1,260	1,260	1,260
R^2	0.013	0.361	0.413

Notes. OLS coefficients; HC1 robust standard errors in parentheses. Average appearance is the reference group. Worker controls include education, experience and its square, female, Black, and good health. Model 3 adds union, married, South, big city, small city, and service industry. Full Model 3 output appears in Appendix A.

The unadjusted below-average appearance coefficient is −0.209. It falls in absolute magnitude to −0.157 after adding worker controls and to −0.133 in the full model. These movements show that part of the unconditional gap is associated with observed differences in workers and jobs. They do not show how much of the remaining gap is discrimination, nor can a block comparison infer the individual contribution of each newly added variable.

In Model 3, below-average appearance is associated with 12.4 percent lower wages on the transformed log-wage scale. The robust standard error is 0.0395, the p-value is 0.0008, and the transformed 95 percent interval is −19.0 to −5.4 percent. The negative association is therefore both economically substantial and statistically distinguishable from zero within this specification. The exact transformation is preferable to the linear approximation when interpreting a coefficient of this magnitude.

The above-average coefficient is 0.012, corresponding to a 1.2 percent premium, with a 95 percent interval from −4.7 to 7.5 percent and p = 0.692. The estimate provides little evidence of a positive premium in this sample, but its uncertainty includes gains that could still matter economically. The appropriate conclusion is that the premium is imprecisely estimated, rather than that above-average appearance has no value in the labor market.

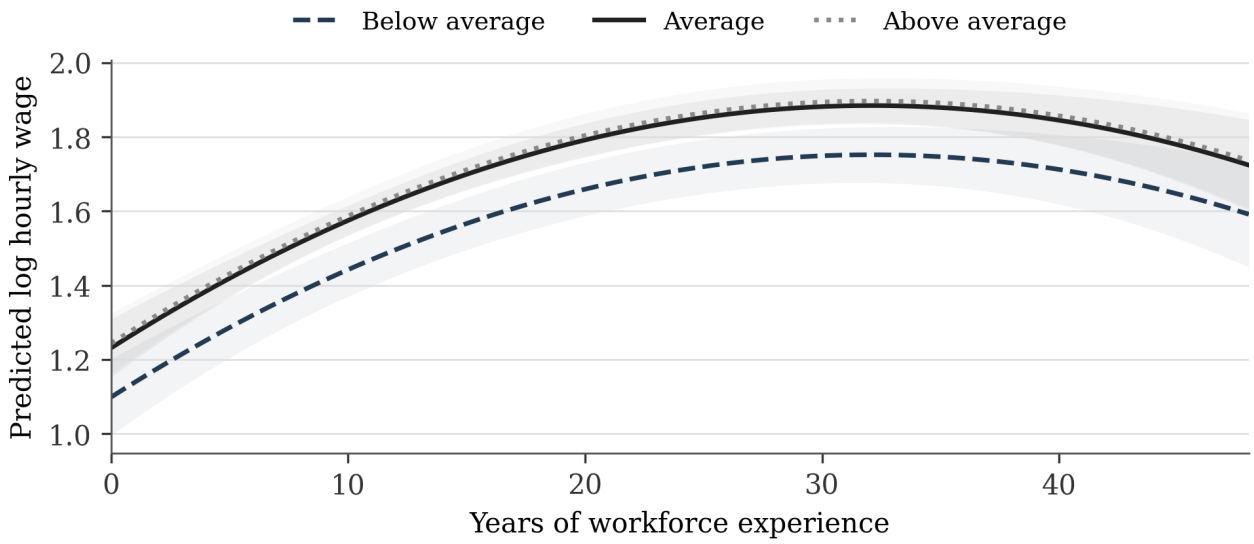
The appearance coefficients are jointly significant (robust F = 6.53, p = 0.0015). The equal-and-opposite restriction is also rejected at the 5 percent level (F = 4.59, p = 0.032). This gives direct statistical support for an imbalance between the estimated penalty and premium. Model 3 explains 41.3 percent of the sample variation in log wages; R^2 measures fit and does not validate a causal interpretation. Education has a positive coefficient, and the experience profile is concave, broadly consistent with a conventional wage equation.

Experience profiles and what they show

Figure 1 reconstructs adjusted predictions from Model 3. At each experience value, we update both experience and experience squared consistently. The remaining covariates are held at their sample

means, so the curves describe standardized predictions for the same covariate profile rather than observed wage trajectories for workers.

Figure 1: *Adjusted log wages over workforce experience*



Notes. Model 3 predictions; shaded regions are pointwise 95 percent normal-approximation intervals using the HC1 covariance matrix. The range is the observed 0–48 years of workforce experience. Other covariates are held at sample means. Average appearance is the reference group.

Predicted log wages rise initially, flatten, and turn downward at approximately 32.1 years of experience. This shape follows from the positive experience coefficient and negative coefficient on its square. It should not be read as an estimate of what happens to the earnings of the same person as that person ages: experience differs across individuals in a single cross section, and the fitted profile can also reflect cohort composition and selection into employment.

Constant log-wage differences vertically separate the three curves because the baseline model contains no interactions between appearance and experience. Their parallel shape is imposed by the specification. It is not independent evidence that employers set an appearance gap at entry and preserve it through promotions, or that the same gap persists over an entire career.

As a limited specification check, we add interactions of both appearance indicators with linear experience while retaining the common quadratic term. The interactions are jointly insignificant

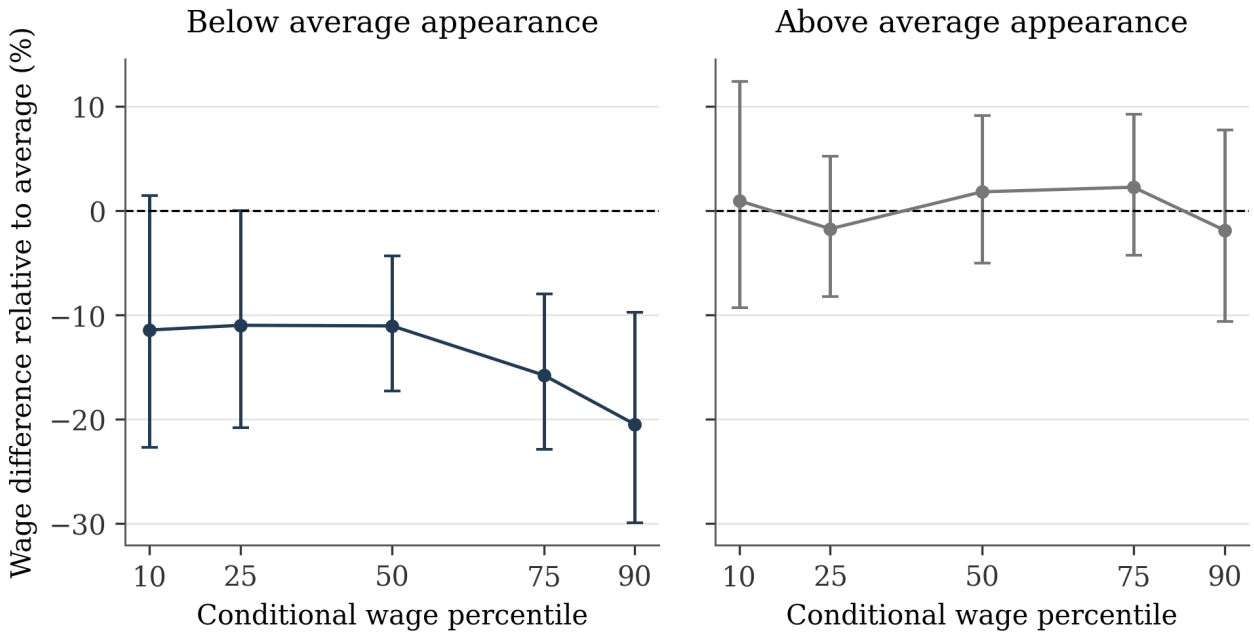
(robust p = 0.954). This provides no evidence of a linear change in the appearance association with experience in this sample. It does not rule out nonlinear heterogeneity or establish stability over time. A credible test of career persistence would require repeated observations of workers and a design that separates experience from cohort effects.

The useful interpretation of Figure 1 is therefore descriptive: the full model associates below-average appearance with lower predicted log wages for a common set of worker characteristics. The figure communicates the model's fitted relationship; it does not supply additional identification beyond the regression.

Differences across conditional wage quantiles

An average wage equation can conceal variation in the relationship between appearance and earnings. Figure 2 reports conditional quantile estimates using the same controls as Model 3. The horizontal axis indexes positions in the conditional wage distribution, not groups formed by sorting workers on their observed wages.

Figure 2: *Appearance contrasts across conditional wage quantiles*



Notes. Points show 100 × [exp(b) − 1]. Bars are transformed pointwise 95 percent normal-approximation intervals using standard errors from 499 paired bootstrap replications. The dashed line marks zero. The reference group is average appearance. Intervals are not adjusted for multiple comparisons.

The below-average appearance coefficient is -0.121 at the 10th percentile and -0.229 at the 90th percentile. These correspond to proportional differences of -11.4 and -20.5 percent, respectively. The estimates are more negative at the upper end, although they do not imply a smooth increase in the penalty at every intermediate quantile. The full numerical estimates and bootstrap standard errors appear in Appendix B.

The estimated 90th-minus-10th percentile contrast for below-average appearance is -0.108 log points, with bootstrap standard error 0.089 and a 95 percent interval of [-0.281, 0.066]. The direct comparison does not reject equality at the 5 percent level ($p = 0.225$). This comparison is more informative about distributional heterogeneity than asking whether each quantile coefficient separately differs from zero. Overlap or non-overlap of individual confidence intervals is not a substitute for testing the difference.

The above-average estimates remain close to zero relative to their uncertainty. Taken together, the quantile results retain the central asymmetry of the mean regression, while the strength of the evidence for a changing penalty must be judged from the interquartile contrast. Tail estimates rely on less information near the relevant conditional quantile and deserve caution in a sample of this size.

These regressions do not follow workers as they move into higher-paying jobs. A coefficient at the 90th conditional percentile describes a difference in that conditional quantile under the specified model. It cannot establish that a particular worker encounters a growing appearance barrier as earnings rise. Without assumptions linking individuals' ranks across counterfactual distributions, the estimates also cannot be interpreted as effects on the same workers at different wage ranks.

Gender differences under a complete interaction model

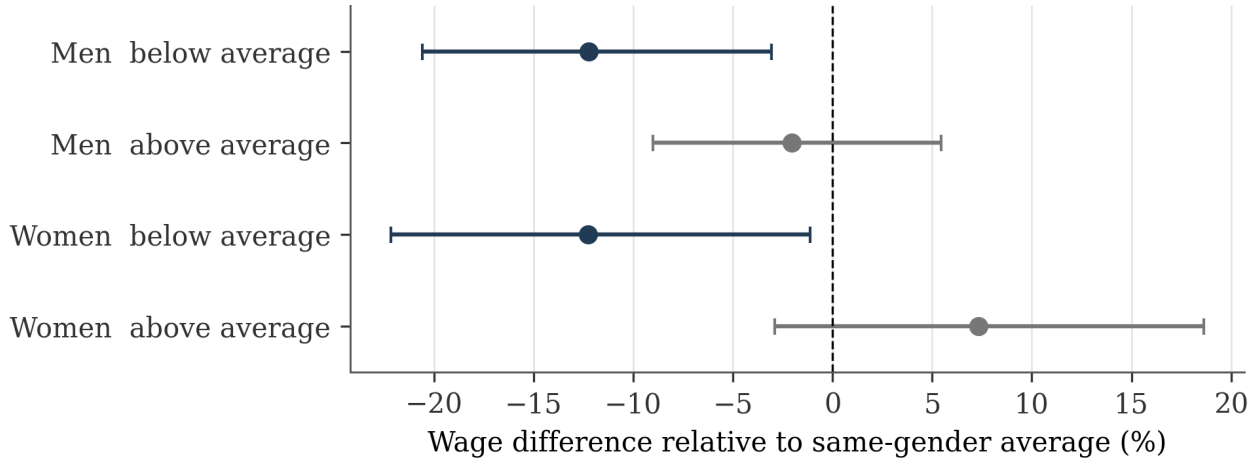
A gender comparison requires a model that allows average-rated women and men to have different baseline wages. We therefore retain the female main effect alongside both appearance

interactions. Omitting it would constrain the adjusted wage difference between average-rated women and men to zero, allowing the interactions to absorb part of the general gender wage gap. Including all lower-order terms separates the within-gender appearance contrasts from the baseline gender difference.

$$\ln(w_i) = \alpha + \beta_B B_i + \beta_A A_i + \delta F_i + \theta_B (B_i F_i) + \theta_A (A_i F_i) + \mathbf{Z}_i' \gamma + \varepsilon_i$$

F denotes the female indicator. Z contains the full control set excluding that indicator, which enters separately. Figure 3 reports appearance contrasts within each gender, with average-rated workers of the same gender serving as the reference group.

Figure 3: *Appearance contrasts within gender*



Notes. Model 3 is extended with a female main effect and both appearance-by-female interactions. Bars are transformed 95 percent t-based confidence intervals using the HC1 covariance matrix. Women's contrasts sum the main appearance and interaction coefficients, including their covariance.

The estimated below-average appearance coefficients are −0.1307 for men and −0.1309 for women. Both imply a wage difference of approximately −12.3 percent relative to average-rated workers of the same gender. The corresponding log-scale robust standard errors are 0.0508 and 0.0610. The female interaction for below-average appearance is −0.0002 (p = 0.998), providing essentially no point-estimate evidence of a larger penalty for women in this specification.

For above-average appearance, the within-gender coefficients are −0.0208 for men and 0.0706 for women, equivalent to −2.1 and 7.3 percent. Neither contrast is statistically distinguishable from zero at the 5 percent level. The two gender interactions are jointly insignificant ($F = 1.15$, $p = 0.318$). The data therefore do not establish gender differences in appearance-related wage associations, and their confidence intervals remain wide enough to include economically meaningful differences.

The evidence supports a below-average appearance disadvantage within both gender groups, rather than a disproportionately large penalty for women. The female main effect captures a separate conditional wage difference among average-rated workers; it should not be relabeled as an appearance penalty. Sparse subgroup counts, especially the 59 below-average-rated women, also limit how sharply gender heterogeneity can be estimated.

Sensitivity and limits of inference

Table 3: *Sensitivity of the appearance coefficients*

Specification	Below avg.	SE	Above avg.	N
Full model with HC1 inference	−0.1326	0.0395	0.0122	1,260
HC3 full	-0.1326	0.0399	0.0122	1,260
No service	-0.1401	0.0399	0.0125	1,260
Exclude top 1 percent wages	-0.1285	0.0381	0.0099	1,247

Notes. Coefficients are in log points. HC1 robust inference is used unless HC3 is indicated. The top-wage exclusion removes observations above the sample 99th percentile; it is a sensitivity exercise, not the preferred sample definition. “No service” retains all other Model 3 controls.

The below-average coefficient changes little when HC3 replaces HC1 inference, when service-industry employment is omitted, or when the highest 1 percent of wages is excluded. The corresponding estimates range from −0.129 to −0.140. These checks suggest that the central result is not driven solely by the particular robust covariance correction, the single service-industry control, or

the extreme upper wage tail. They address a limited set of statistical concerns and do not resolve omitted-variable bias.

First, education and experience are incomplete measures of productivity. Ability, parental resources, interpersonal skills, job tenure, detailed occupation, and employer characteristics are unavailable. If they are related to both appearance ratings and wages, the appearance coefficients combine their influence with other channels. Robust standard errors correct a form of inference under heteroskedasticity; they cannot correct a misspecified conditional mean or remove confounding.

Second, job characteristics can be outcomes of the same process being studied. If appearance affects access to certain jobs, controlling for industry can remove part of the overall association. If unobserved skills influence both job allocation and pay, conditioning on employment characteristics may introduce further selection. The similarity of the estimates with and without service employment is informative about this coarse variable, but it cannot establish that occupational sorting is unimportant.

Third, appearance is measured through subjective interviewer judgments. Ratings may contain random error or systematic differences across interviewers and social groups. They may also respond to grooming or presentation that depend on income. The direction of bias is therefore not known in general. A simple attenuation argument for classical error in a continuous regressor does not automatically apply to misclassified appearance categories.

Fourth, wages are observed for employed respondents. If appearance is related to participation or hiring, the selected wage sample may differ systematically from the population of potential workers. This concern is especially relevant when interpreting gender comparisons. The extract lacks the excluded variables and nonworker information needed to implement and defend a selection correction. We therefore make no claim that the estimated wage gap describes the total economic consequence of appearance.

Finally, the extract contains no survey-design identifiers for clustered or design-based inference. HC1 and the paired bootstrap treat worker records as independent observations. All extensions are exploratory, and the quantile intervals are pointwise. Multiple specifications and comparisons make isolated significance thresholds less persuasive than a stable pattern, transparent uncertainty, and an explicit account of the sample's limits.

Discussion and conclusion

The strongest result is a negative conditional wage association for below-average appearance. It is visible before controls, remains after substantial adjustment, and survives the limited sensitivity checks considered here. The full-model coefficient implies a 12.4 percent difference on the transformed log-wage scale. Above-average appearance has a much smaller and imprecisely estimated coefficient, and a direct test rejects equal-and-opposite penalty and premium magnitudes. This is a more defensible basis for describing asymmetry than the significance of the penalty alone.

The result broadly agrees with the original study's emphasis on a below-average appearance penalty. Its magnitude need not match every published estimate because the teaching extract retains fewer variables and does not reproduce all of the original study's sample restrictions. A larger coefficient in this exercise should not be interpreted as evidence that appearance bias has worsened over time. There is no contemporary comparison sample and no repeated design capable of identifying such a trend.

The extensions clarify both the scope and the limits of the evidence. The experience curves show the fitted quadratic wage profile, but their parallel shape follows from an additive model. The quantile estimates describe conditional distributional differences and do not track career advancement. The full gender interaction model yields nearly identical below-average appearance-penalty estimates for men and women, with no statistically detectable interaction.

Whether the residual association reflects discrimination remains an open question. Employer preferences and biased beliefs are plausible, but so are unobserved interpersonal skills, presentation,

and selection into jobs. Even a channel that increases customer demand would not settle the normative question of whether the resulting treatment is fair. Economic productivity, unequal treatment, and social evaluation are distinct concepts, and this dataset does not allow them to be collapsed into a single explanation.

The most valuable next step is a design that can distinguish mechanisms. A contemporary correspondence or hiring experiment could randomize appearance signals while holding qualifications constant, subject to ethical review and realistic treatment construction. Such a study would identify effects at the hiring stage, rather than automatically identifying wage effects among incumbent workers. Repeated worker observations with appearance assessments, detailed occupations, employer information, and participation outcomes would address different questions about sorting and career development.

For distributional analysis, a larger sample and more extensive bootstrap inference would improve precision, particularly in the tails and within appearance-by-gender groups. A pre-specified set of quantiles and contrasts would also reduce the risk of emphasizing whichever result happens to cross a conventional significance threshold. Better measurement would include multiple raters, information on rating conditions, and a clearer distinction between relatively stable facial characteristics and income-related grooming or presentation.

This replication supports a bounded but economically meaningful conclusion: in the historical beauty sample, workers rated below average in appearance receive lower wages than observably comparable workers rated average, while the premium for above-average appearance is less clearly established. It does not demonstrate a contemporary trend, a career-long causal penalty, or a larger penalty for women. Recognizing those boundaries makes the result more credible and clarifies the evidence needed to explain why the wage association exists.

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Reproducibility and scope

All descriptive statistics, regressions, tests, and figures in this paper were recomputed from the public CSV copy of beauty. The analysis checks the sample dimensions, missingness, appearance-category coding, and consistency of experience squared. It also verifies that the full OLS specification reproduces the original baseline appearance coefficients and R^2. Matching these results supports numerical comparability; it is not a claim that the original authors' complete data construction has been reproduced.

OLS coefficients are obtained by least squares. HC1 covariance estimates apply the finite-sample multiplier $N/(N - K)$, where K includes the intercept. OLS tests use the residual degrees of freedom. HC3 is reported as a sensitivity check. Quantile coefficients are solved by linear programming, minimizing the standard quantile check loss. The squared-experience column is rescaled for numerical conditioning without changing the model. Paired bootstrap replications resample worker records, retaining the same resample across all five quantiles. Interquantile inference uses the bootstrap standard deviation of the coefficient difference.

The companion replication archive contains the public data copy, analysis code, numerical output, and figure files. The code records the source URL, data checksum, bootstrap seed, and number of replications. Appendices A and B report numerical results behind the main presentation. The code implements the complete gender interaction specification and updates the quadratic experience term consistently when constructing predictions.

Appendix A: Full mean regression

Table A1: Model 3 estimates

Variable	Coefficient	Robust SE	p value	95 percent CI
Intercept	0.3230	0.1029	0.002	[0.1212, 0.5249]
Below-average appearance	-0.1326	0.0395	<0.001	[-0.2101, -0.0550]
Above-average appearance	0.0122	0.0307	0.692	[-0.0481, 0.0725]
Education	0.0673	0.0056	<0.001	[0.0563, 0.0783]
Experience	0.0407	0.0042	<0.001	[0.0326, 0.0489]
Experience squared	-0.0006	0.0001	<0.001	[-0.0008, -0.0005]
Union member	0.1602	0.0277	<0.001	[0.1059, 0.2145]
Black indicator	-0.0849	0.0588	0.149	[-0.2001, 0.0304]
Female indicator	-0.3804	0.0311	<0.001	[-0.4414, -0.3195]
Married	0.0364	0.0306	0.235	[-0.0236, 0.0964]
Good health	0.0689	0.0633	0.277	[-0.0553, 0.1931]
South	0.0682	0.0321	0.034	[0.0053, 0.1311]
Big city	0.2391	0.0375	<0.001	[0.1656, 0.3127]
Small city	0.0903	0.0308	0.003	[0.0299, 0.1506]
Service industry	-0.1445	0.0335	<0.001	[-0.2103, -0.0788]

Notes. Dependent variable: log hourly wage. N = 1,260; $R^2 = 0.4131$. HC1 heteroskedasticity-robust standard errors; t-based intervals and two-sided p values. Average appearance is the omitted appearance category. City indicators are relative to neither big nor small city. The female and Black indicators retain the source dataset's coding.

Appendix B: Conditional quantile estimates

Table B1: Appearance coefficients by conditional quantile

Percentile	Below avg.	Bootstrap SE	Above avg.	Bootstrap SE
10	-0.1214	0.0693	0.0094	0.0548
25	-0.1164	0.0594	-0.0176	0.0349
50	-0.1170	0.0371	0.0179	0.0354
75	-0.1715	0.0451	0.0222	0.0337
90	-0.2290	0.0645	-0.0189	0.0478

Notes. All estimates include the Model 3 controls. Coefficients are in log points; 499 paired bootstrap replications, seed 20261105. Figure 2 displays transformed point estimates and intervals. The 90th-minus-10th below-average coefficient difference is -0.1076 (bootstrap SE 0.0887; two-sided normal-approximation p = 0.225).